DIGITAL ACQUISITION AND IMAGING REPORT

Acquisition and Analysis of a Formatted 8 GB USB Flash Drive

Case number	001
Exhibit number	Img 01
Examiner	El-sly Antwi
Course / module	Digital Acquisition and Imaging
Report date	13 October 2026
Evidence	8 GB USB flash drive (reported SanDisk), image file MyDrive.E01
Tools	FTK Imager; Autopsy 4.22.1
Version	1.0

Prepared for course assessment. Findings are limited to the evidence shown in the figures.

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1. Executive Summary

An 8 GB USB flash drive (reported by the examiner as a SanDisk device; presented to FTK Imager as "SMI USB DISK USB Device") that had been quick-formatted to FAT32 was forensically imaged and examined. The drive was acquired with FTK Imager as an E01 image, and the image was verified: the computed, stored and report SHA1 hashes are identical and no bad blocks were found.

The image was then analysed in Autopsy 4.22.1. Although the drive had been formatted, Autopsy listed 48 deleted items (25 file-system entries and 23 carved items), 15 orphan files and 8 unallocated-space segments. Eighteen files were classified as video by MIME type (12 Matroska and 6 MPEG), and a built-in keyword search returned 44 email-address hits.

In plain terms: a quick format removes the drive's directory structure but does not erase the underlying data, and traces of earlier content remain recoverable. The recovered content has **not** been opened or reviewed, so this report makes no claim about what the files contain, who created them, or why the drive was formatted.

2. Scope and Objectives

This report documents the forensic acquisition and analysis of one USB flash drive (Exhibit Img 01, Case 001). The objectives were to:

1. acquire a forensically sound image of the device and prove its integrity by hash verification;
2. describe the file-system structure of the image;
3. identify deleted, orphaned and carved data that remains despite the quick format; and
4. report findings so that observed facts are kept separate from interpretation.

Out of scope: review of the content of recovered media, attribution to a person, and any determination of intent. These are noted as recommendations in Section 9.

Reporting conventions. Each finding is split into an "Observation" (directly visible in a cited figure) and an "Assessment" (the examiner's interpretation). In the figures, **red boxes** mark the items relied on, and **blurred areas** redact personal information (the Windows account name in file paths and personal folder names). Hash values, case details and device details are left readable because they are evidence.

3. Evidence Handled

3.1 Device description

Attribute	Detail
Exhibit / case	Exhibit Img 01; Case 001 (Autopsy). The FTK Imager case field was entered as "Imaginng 001" and is quoted as entered.
Device	8 GB USB flash drive; make reported by the examiner as SanDisk
Name shown by FTK Imager	SMI USB DISK USB Device [8GB USB], \\.\PHYSICALDRIVE1 (Figure 4)
Capacity (Windows)	7.61 GB total; 7.60 GB shown in Explorer (Figure 1)
File system after format	FAT32, 4096-byte allocation units, volume label "NO NAME", Quick Format (Figure 2)
Examiner	El-sly Antwi

Note on identification. FTK Imager reports the device by its controller string ("SMI USB DISK"), which does not necessarily match the brand printed on the casing. The SanDisk make is therefore recorded as examiner-reported and should be confirmed against the physical label.

3.2 State of the device before acquisition

The drive was quick-formatted to FAT32 before it was imaged. Before the format, Windows showed drive D: with 6.51 GB free of 7.60 GB, i.e. about 1.09 GB in use (Figure 1). After the format it showed 7.60 GB free of 7.60 GB (Figure 2). The date and time of the format were not captured. Because the format changed the device before acquisition, the original state of the drive cannot be established by this examination.

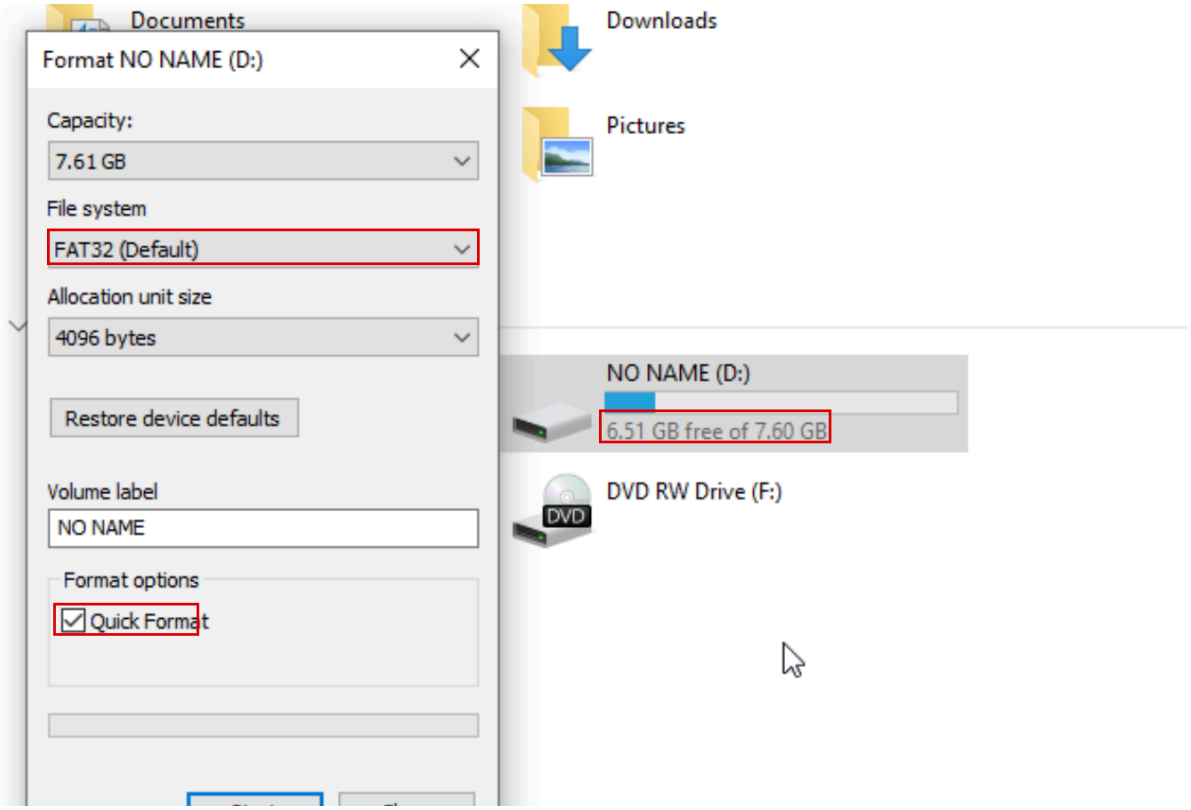


Figure 1: *Windows Format dialog for drive D: before formatting (FAT32, 4096-byte allocation unit, Quick Format ticked; 6.51 GB free of 7.60 GB).*

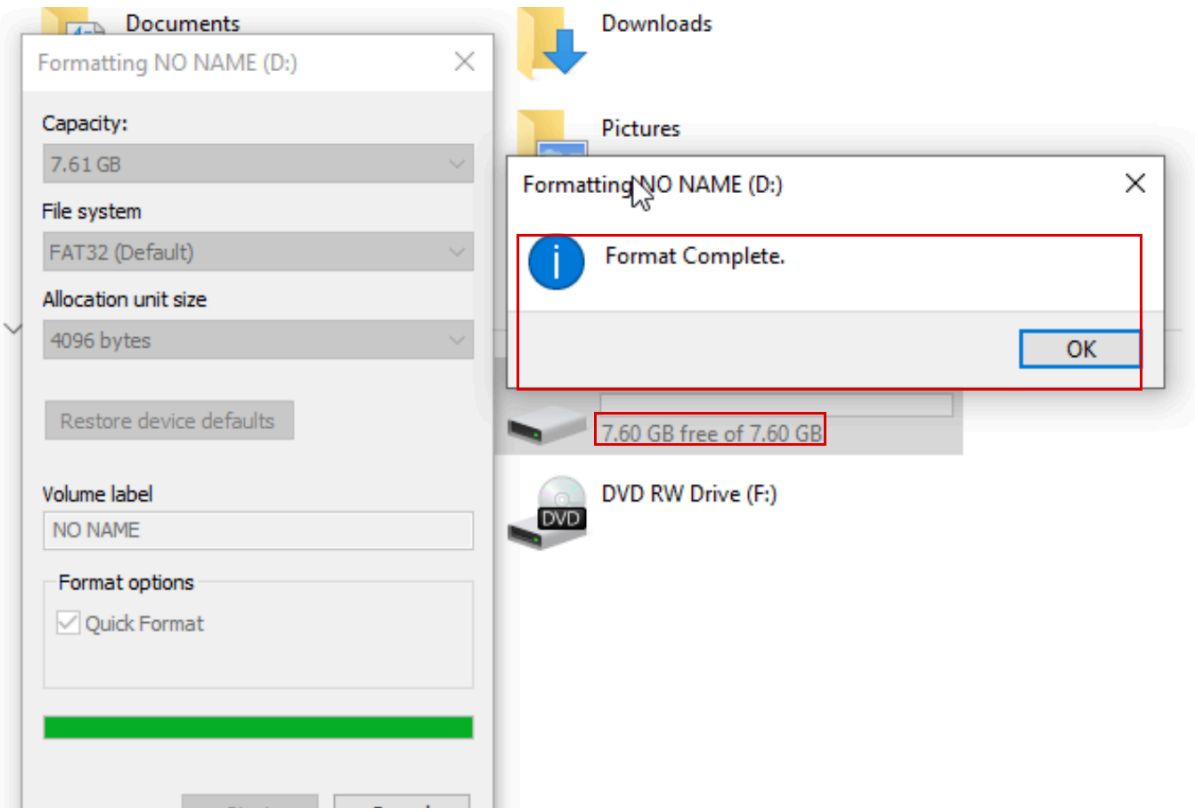


Figure 2: *Format completed; drive D: now shows 7.60 GB free of 7.60 GB.*

3.3 Chain of custody

#	Date / time	Action	Handler	Reference
1	Not captured	Drive quick-formatted to FAT32 (label NO NAME) before acquisition	Not recorded	Figure 1, Figure 2
2	Not captured	Physical drive imaged with FTK Imager to MyDrive.E01 (duration 0:09:14)	El-sly Antwi	Figures 3 to 11
3	Not captured	Image verified: SHA1 computed, stored and report hashes match	El-sly Antwi	Figure 12
4	Not captured	Autopsy case "My Drive Forensics" created; MyDrive.E01 added as data source	El-sly Antwi	Figures 13 to 16
5	Not captured	Ingest modules run and results reviewed	El-sly Antwi	Figures 17 to 25
6	13 Oct 2026, 17:08:32 (host local time, from report folder name)	Autopsy HTML report generated	El-sly Antwi	Figure 26

Receipt and storage details for the original device were not captured in the screenshots and should be added by the examiner if the submission requires them.

4. Tools and Environment

Item	Detail
Acquisition tool	FTK Imager, version 8.3.0.27
Analysis tool	Autopsy 4.22.1 (shown in the title bar, Figure 18)
Host operating system	Windows 10
Write blocker	None used (see Section 8)

Item	Detail
Image format	E01 (EnCase), compression level 6, fragment size 1500 MB
Time zone for analysis	UTC (GMT+0:00); all Autopsy timestamps in this report are UTC
Image file	MyDrive.E01

Note. No write blocker was used. The integrity of the evidence therefore relies on the hash verification in Section 5.4, which confirms that the image matches what was read, not that the source was unmodified during acquisition.

5. Acquisition (FTK Imager)

5.1 Source selection

The Physical Drive option was selected (Figure 3), and \\.\PHYSICALDRIVE1, the "SMI USB DISK USB Device [8GB USB]", was chosen as the source (Figure 4). The other listed disk, PHYSICALDRIVE0, is the examiner's 500 GB internal drive and was not imaged. Imaging the physical drive rather than the logical volume captures unallocated space and the partition area, which is essential for recovering deleted data.

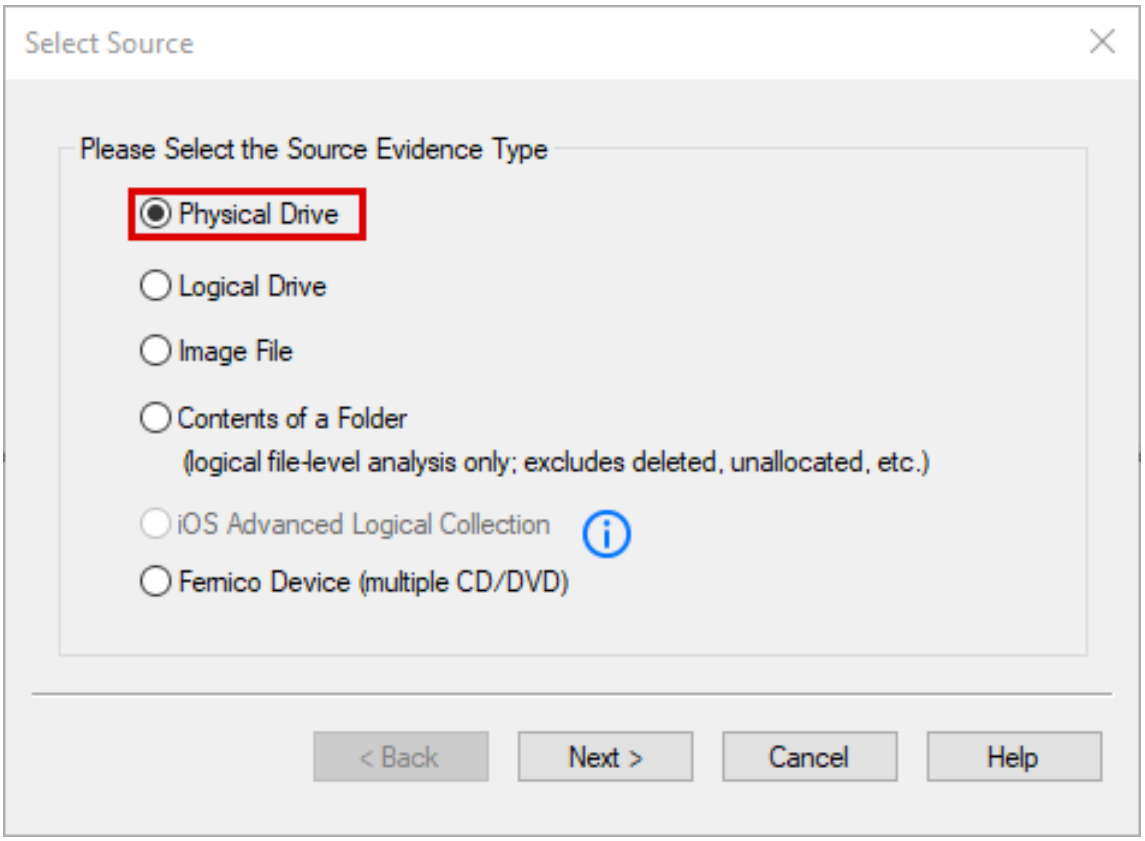


Figure 3: *FTK Imager, Select Source: Physical Drive chosen as the evidence type.*

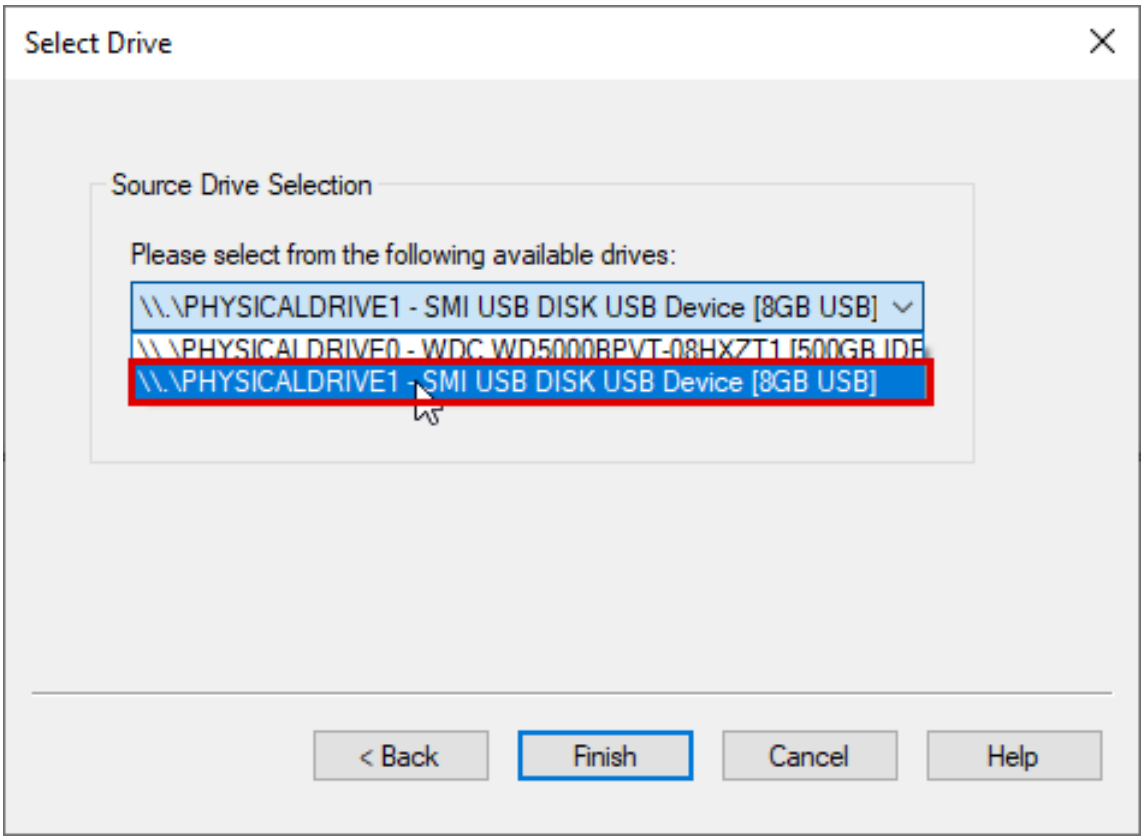


Figure 4: *FTK Imager, Select Drive: \\.\PHYSICALDRIVE1 (SMI USB DISK USB Device, 8 GB) selected.*

5.2 Image type and evidence details

E01 was selected as the image type (Figure 5). Evidence Item Information was entered as case number "Imaginng 001", evidence number "Img 01" and examiner "El-sly Antwi"; the description and notes fields were left empty (Figure 6).

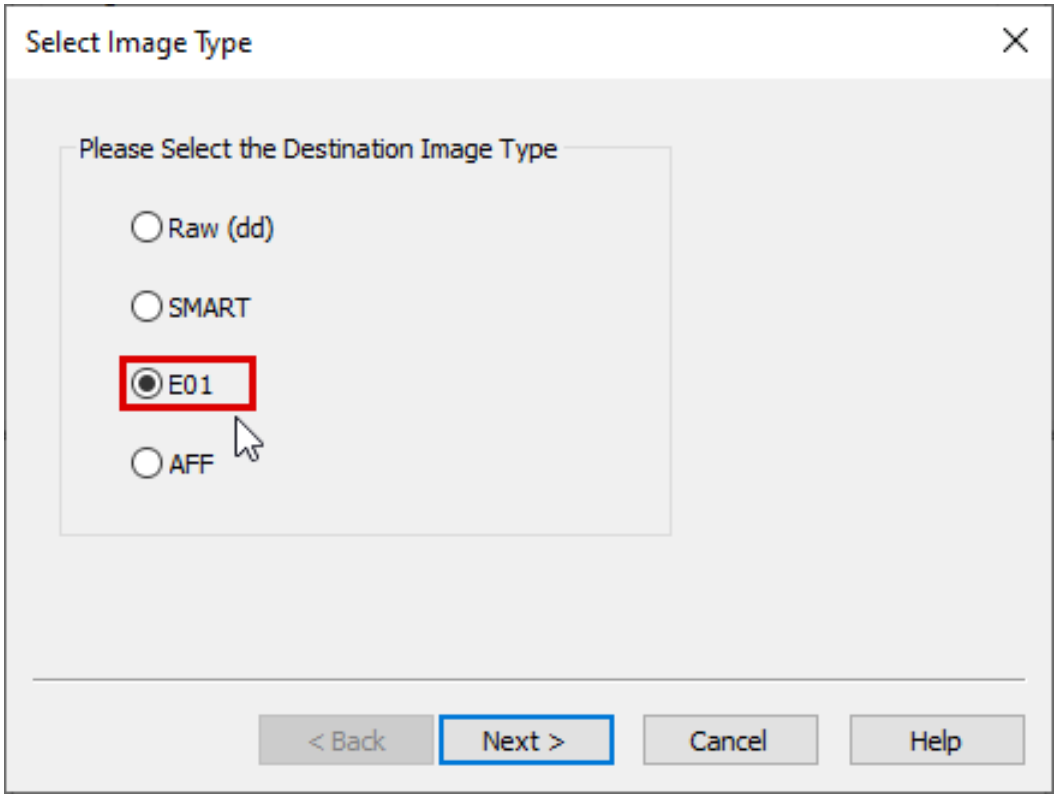


Figure 5: *FTK Imager, Select Image Type: E01 chosen.*

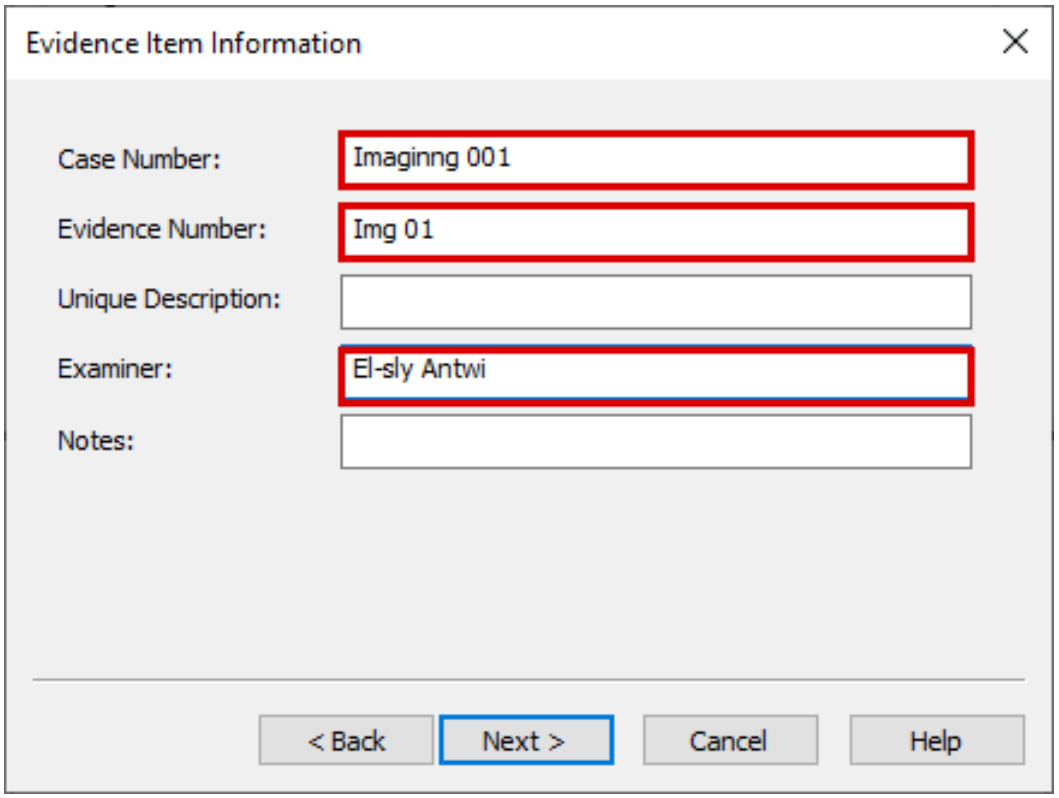


Figure 6: *FTK Imager, Evidence Item Information: case number, evidence number and examiner as entered.*

5.3 Destination and settings

A destination folder named "Drive IMG" was selected (Figure 7). Image fragment size was 1500 MB, compression level 6, and AD encryption was not used (Figure 8). The summary shows the destination as "Evidence\MyDrive [E01]" and "Verify images after they are created" ticked (Figure 9).

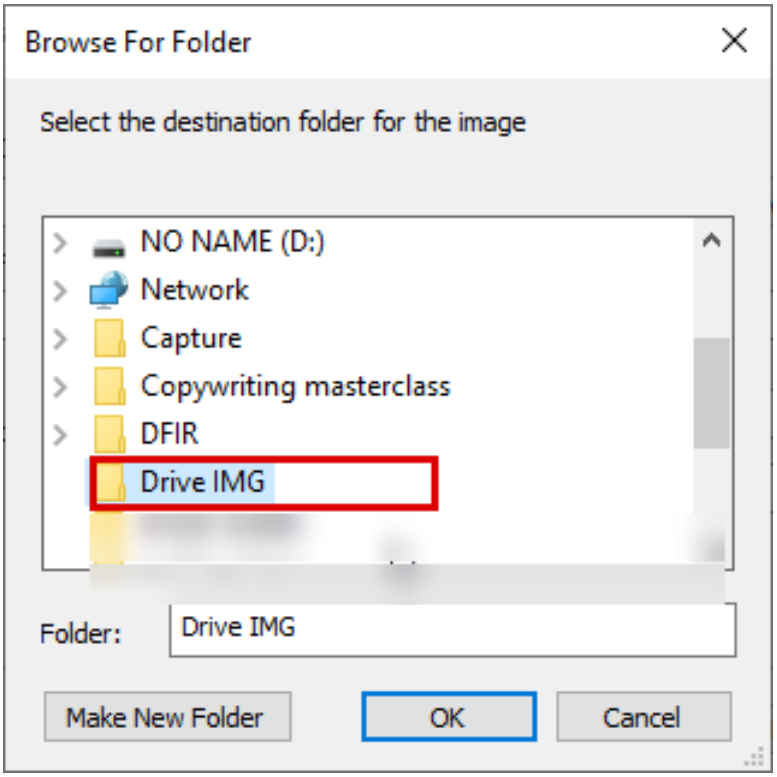


Figure 7: *Browse For Folder dialog used to choose the destination folder (personal folder names blurred).*

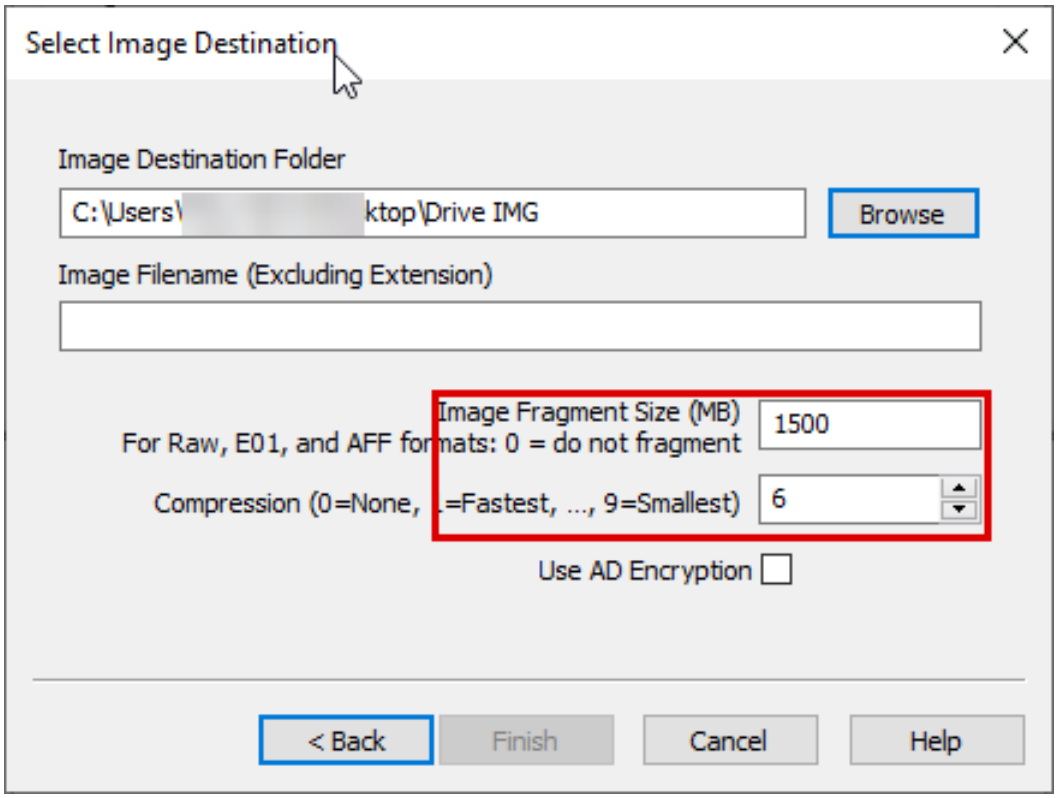


Figure 8: *Select Image Destination: fragment size 1500 MB, compression 6, AD encryption not used (username blurred).*

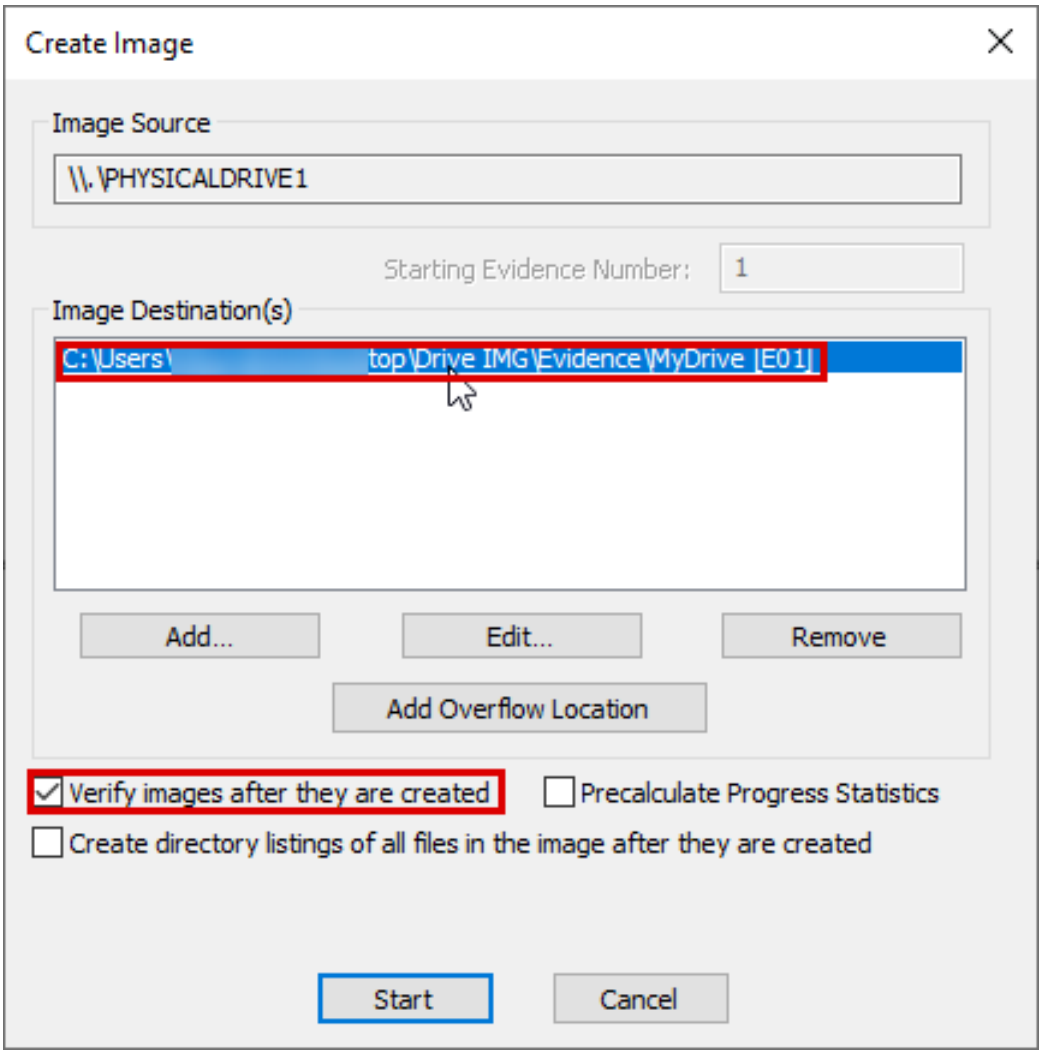


Figure 9: *Create Image summary: source \\.\PHYSICALDRIVE1, destination MyDrive [E01], "Verify images after they are created" ticked (username blurred).*

5.4 Imaging and verification

Imaging started (Figure 10) and finished with the status "Image created successfully" after 0:09:14 (Figure 11). The verification window (Figure 12) reports:

- SHA1 computed hash, stored verification hash and report hash are all **bab4ed750a1e4cb820028fc3b257e19857a8c73b**, with result **Match**;
- a further "Verify result: Match" row, visible at the top of the window, belongs to the section above the SHA1 block (the MD5 values were scrolled out of view); and
- "No bad blocks found in image".

MD5 values were not recorded during the examination (see Section 8); SHA1 is the only hash value relied on in this report.

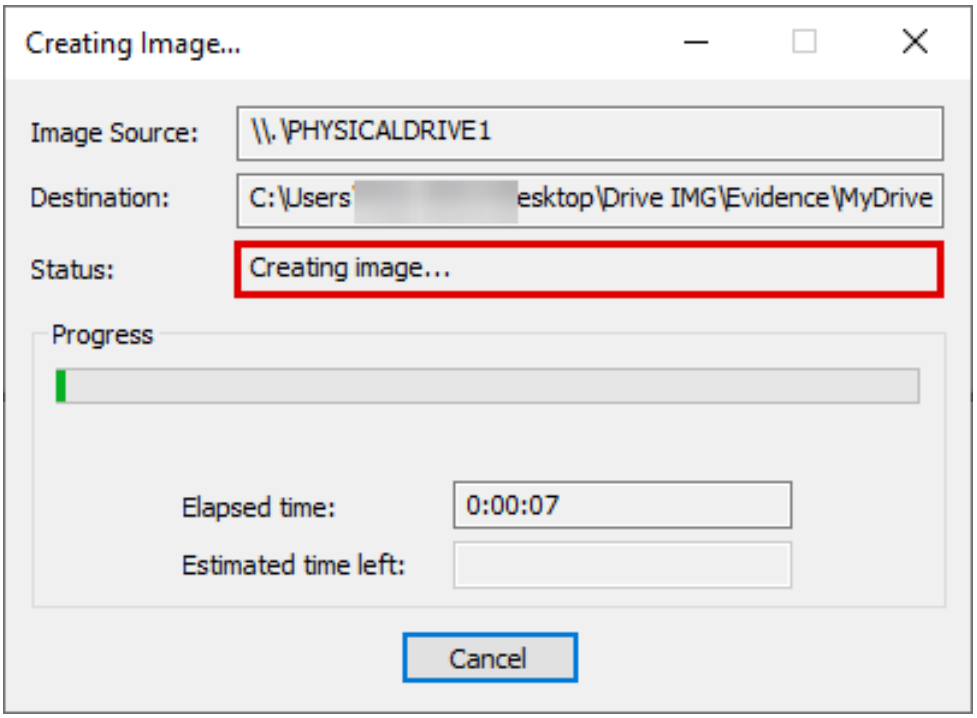


Figure 10: *Imaging in progress (elapsed 0:00:07; username blurred).*

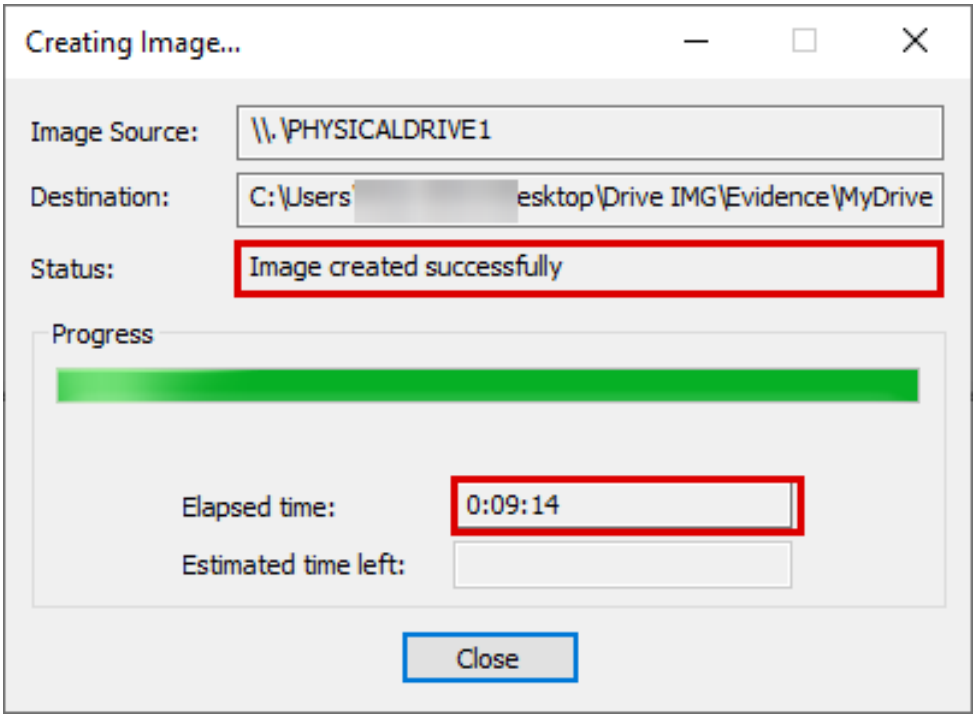


Figure 11: *Imaging complete: "Image created successfully", elapsed time 0:09:14 (username blurred).*

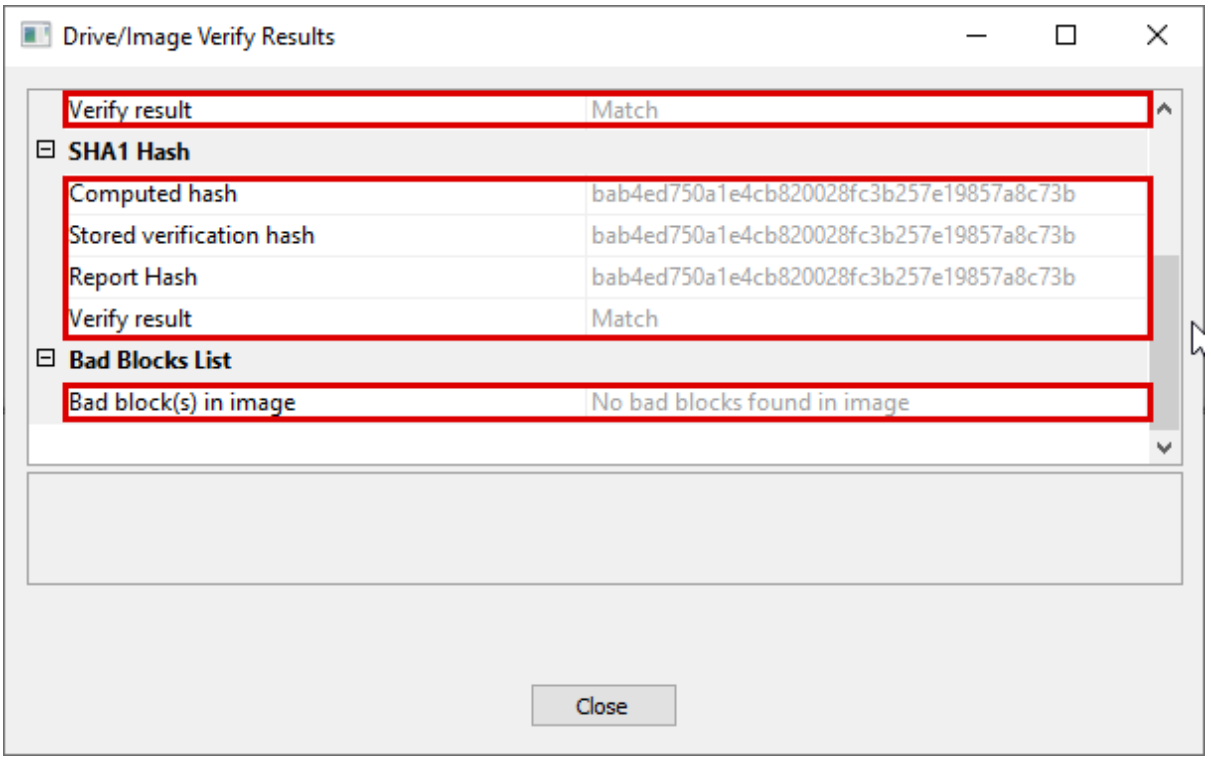


Figure 12: *Drive/Image Verify Results: SHA1 computed, stored and report hashes identical; result Match; no bad blocks.*

6. Analysis (Autopsy)

6.1 Case creation and data source

A single-user case named "My Drive Forensics" was created, with case number 001 and examiner El-sly Antwi (Figure 13, Figure 14). The data source type "Disk Image or VM File" was chosen (Figure 15) and MyDrive.E01 was added with the time zone set to UTC and sector size on auto-detect (Figure 16). The option "Ignore orphan files in FAT file systems" was not ticked, so orphan files were included. Autopsy did not validate the image hashes on import because the optional hash fields were left empty; image integrity was established in FTK Imager (Section 5.4).

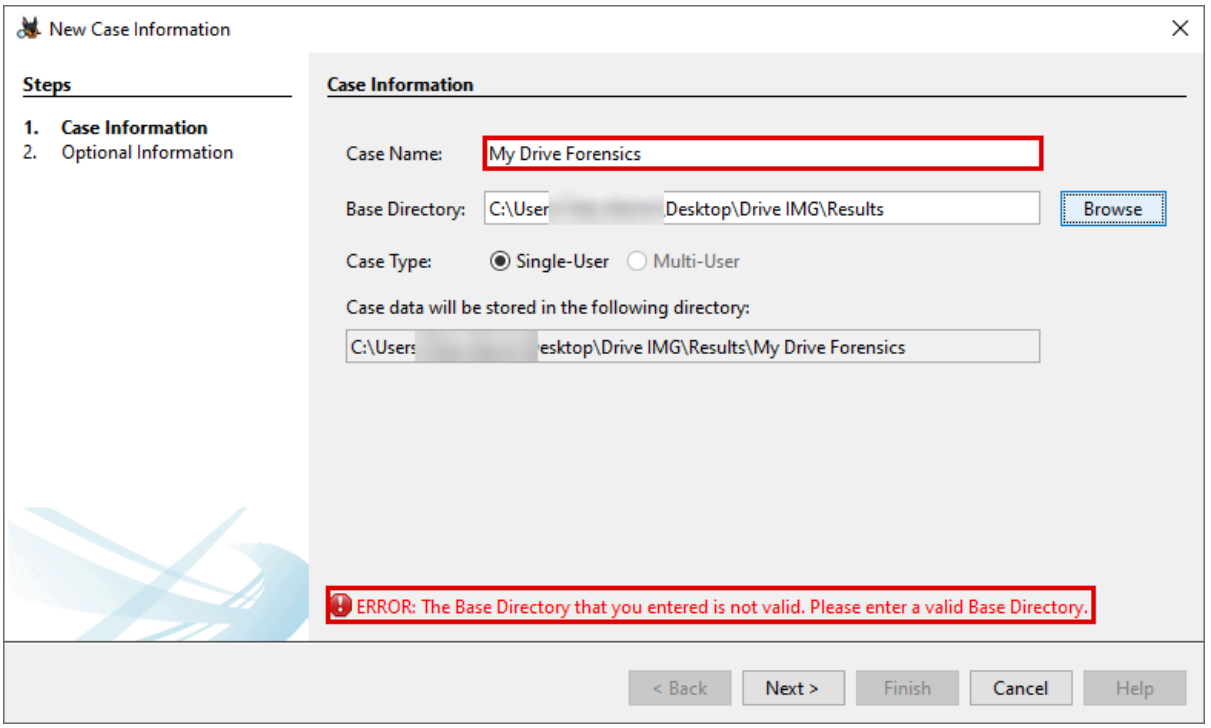


Figure 13: *Autopsy, New Case Information: case name "My Drive Forensics", single-user case (username blurred). The red error line is a transient message visible at the moment of capture.*

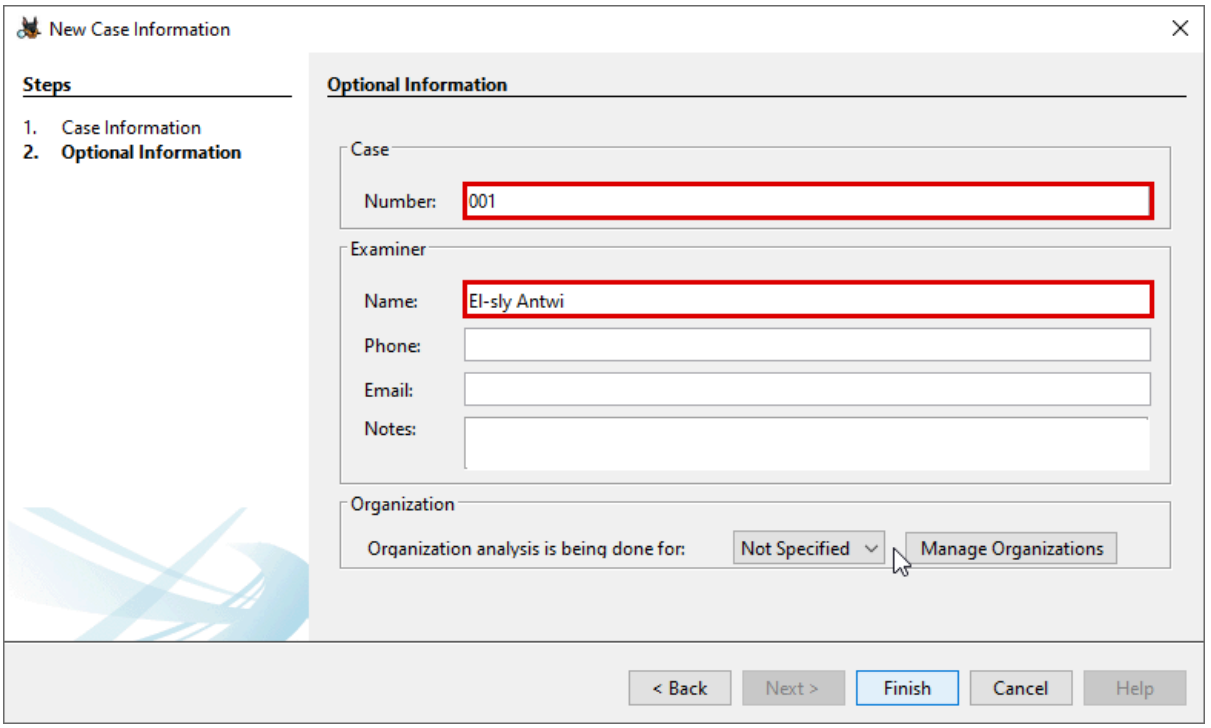


Figure 14: *Autopsy, Optional Information: case number 001 and examiner name.*

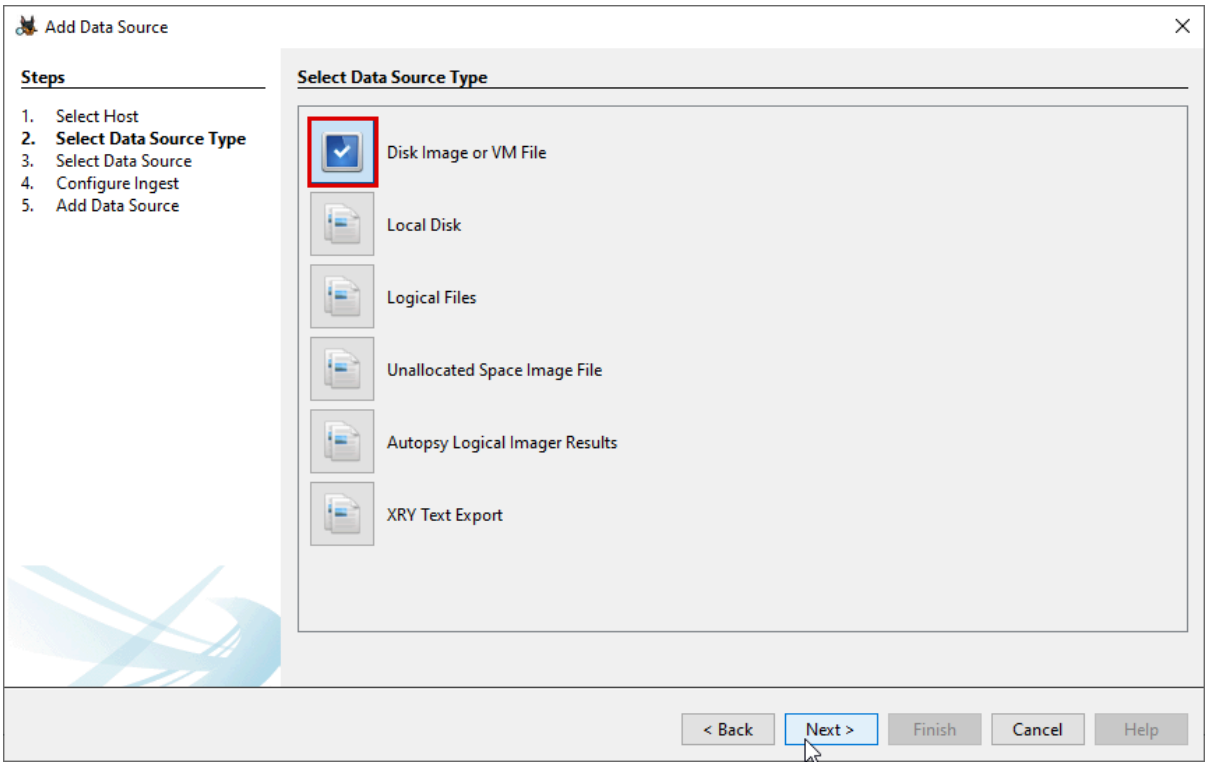


Figure 15: *Autopsy, Select Data Source Type: Disk Image or VM File.*

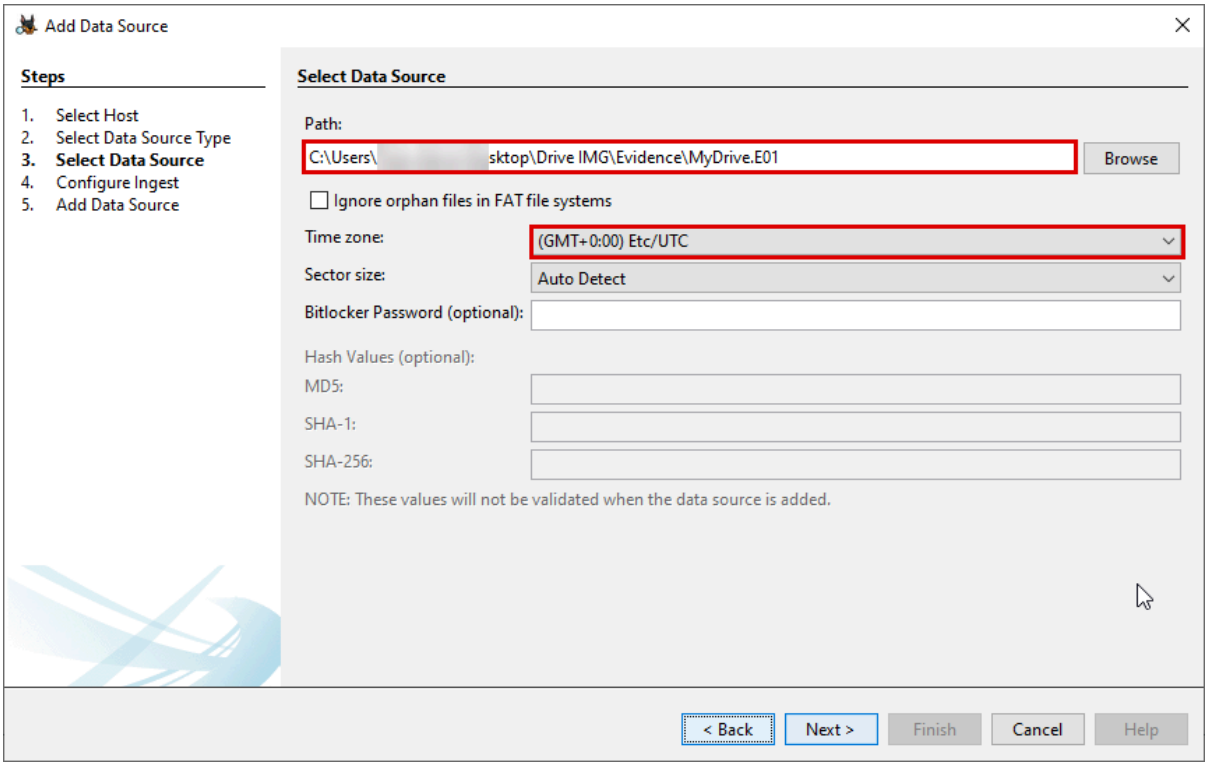


Figure 16: *Autopsy, Select Data Source: MyDrive.E01, time zone UTC (GMT+0:00), sector size auto-detect (username blurred).*

6.2 Ingest modules

Modules ran on "All Files, Directories, and Unallocated Space" (Figure 17). The ticked modules visible in the screenshot were Embedded File Extractor, Picture Analyzer, Keyword Search, Email Parser, Encryption Detection, Interesting Files Identifier, Central Repository, PhotoRec Carver, Virtual Machine Extractor and Data Source Integrity. Android Analyzer (aLEAPP), Cyber Triage Malware Scanner and DJI Drone Analyzer were not ticked. Modules above the visible part of the list were not captured. Autopsy 4.22.1 processed the image (Figure 18).

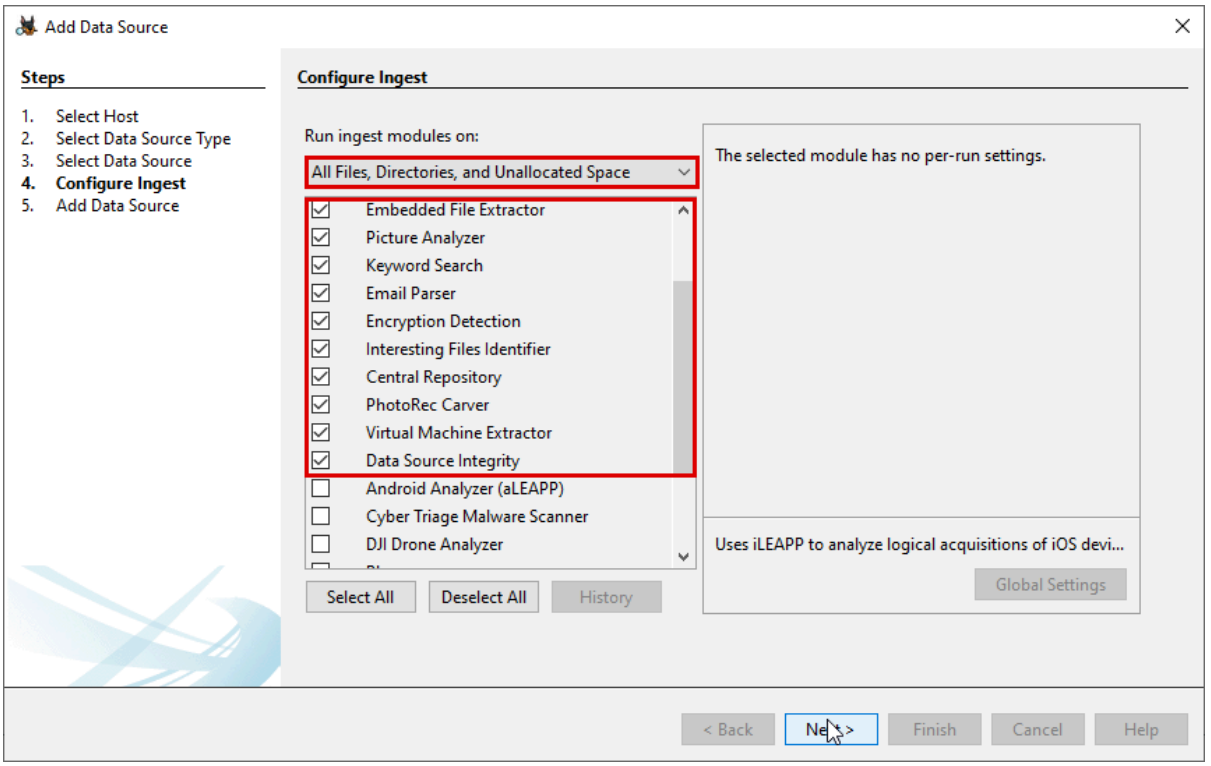


Figure 17: *Autopsy, Configure Ingest: modules run on all files, directories and unallocated space.*

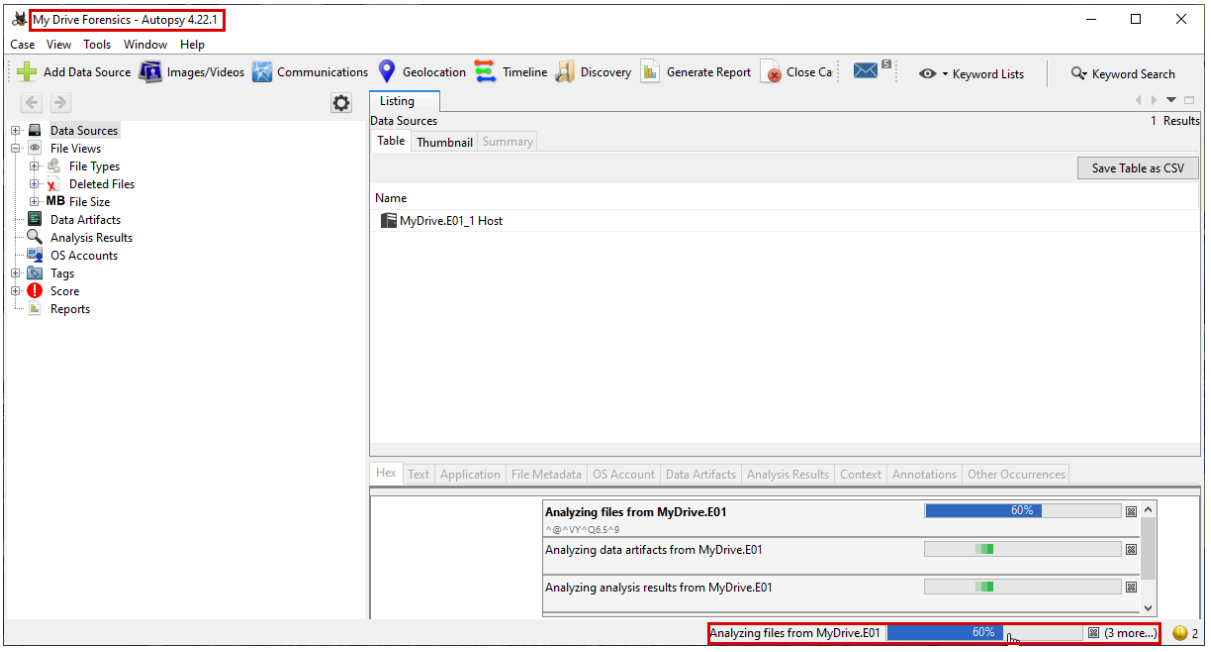


Figure 18: *Autopsy 4.22.1 main window while ingest was running (Analyzing files from MyDrive.E01, 60%).*

6.3 File system structure

The data source MyDrive.E01 contains four special or system folders (Figure 19): $OrphanFiles (15), $CarvedFiles (1), which holds one sub-folder "1" with 23 items, $Unalloc (8) and System Volume Information (4). The $Unalloc folder holds eight unallocated-space segments, the largest of which are about 1 GB each (Figure 20).

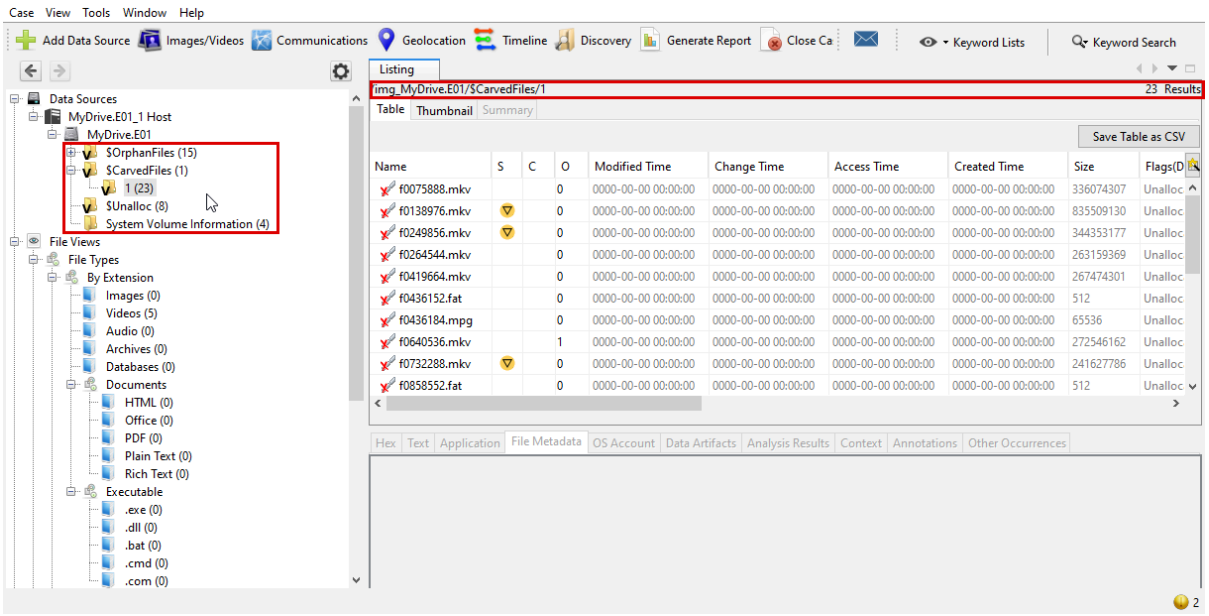


Figure 19: *Data source tree for MyDrive.E01: $OrphanFiles (15), $CarvedFiles (1) containing folder "1" (23), $Unalloc (8), System Volume Information (4).*

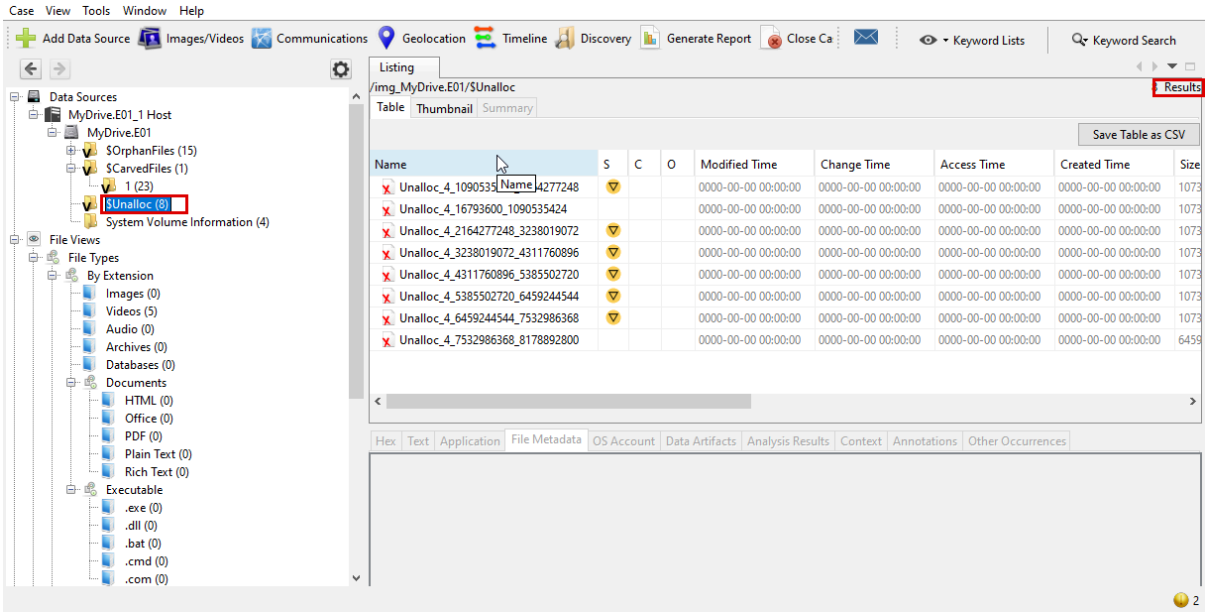


Figure 20: *Contents of $Unalloc: eight unallocated-space segments.*

6.4 Deleted files

Autopsy lists 48 deleted items, of which 25 are file-system entries and the remaining 23 match the number of items under $CarvedFiles/1 (Figure 21). The first ten file-system entries visible include three groups of names (000~1.TS, 000~1.DVR, INFO3~1.DVR), directory placeholders, and an entry named "_TS_20~1.31-". Selecting 000~1.DVR opens the Extracted Text pane (Figure 22); the content shown was not interpreted.

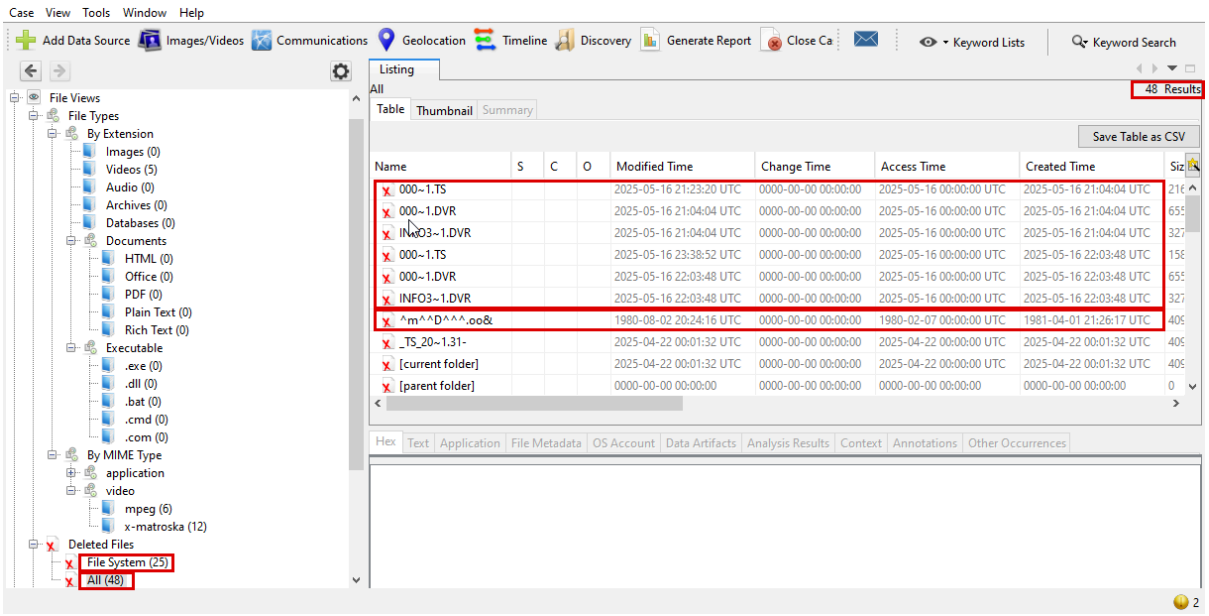


Figure 21: *Deleted Files > All (48): file-system entries with dates; the entry with 1980/1981 timestamps is boxed.*

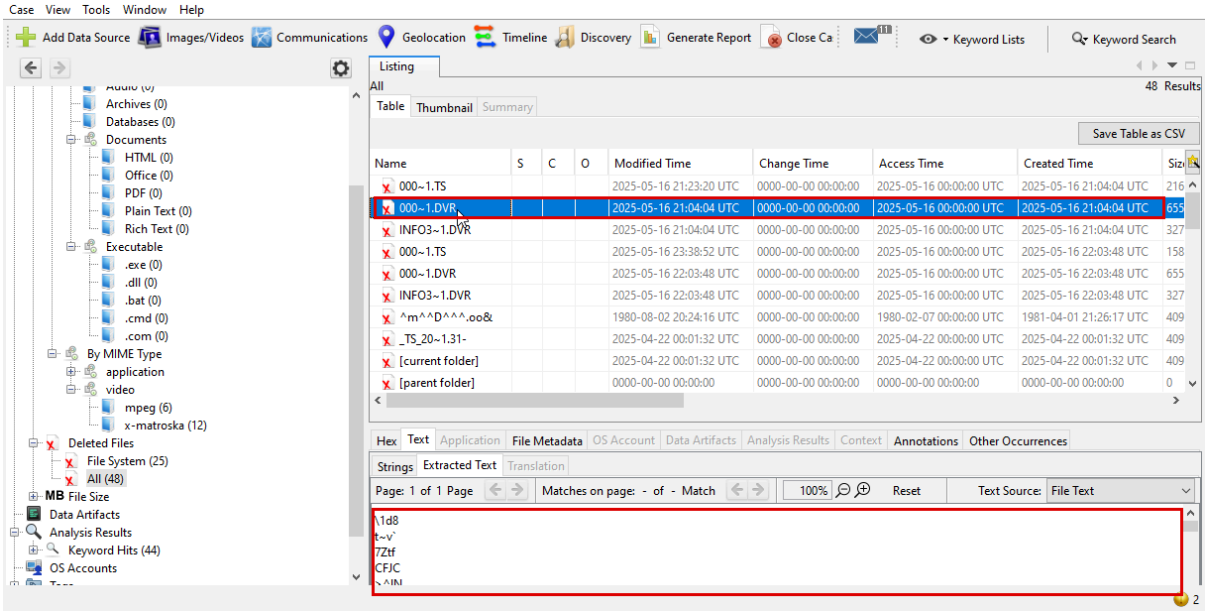


Figure 22: *Deleted entry 000~1.DVR selected, with the Extracted Text pane open.*

6.5 File types

By extension, Autopsy reports 5 videos: one .swf item (231,525,888 bytes) and four .mpg items of 65,536 bytes each, all flagged as unallocated (Figure 23). By MIME type, 18 files are video: 12 x-matroska and 6 mpeg (Figure 24). The visible x-matroska items carry names in the carved-file pattern (f<offset>.mkv) and have no recorded timestamps. Several items carry a yellow marker in the S (score) column; the reason for the marker was not captured. A signature-mismatch view was not captured.

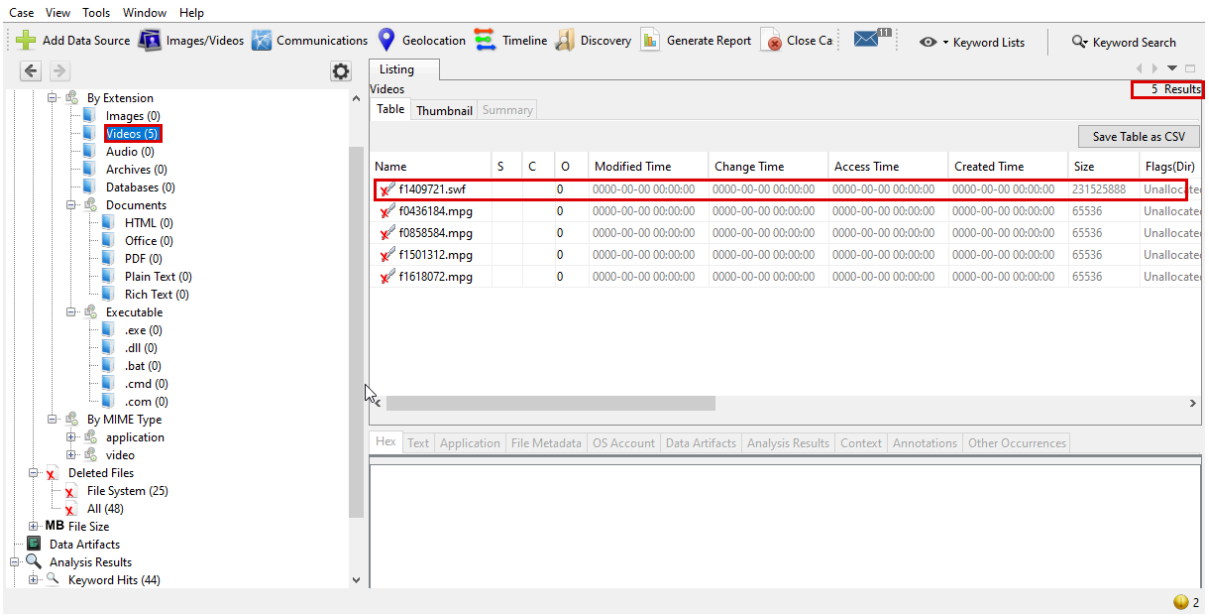


Figure 23: *File Types > By Extension > Videos (5), including one .swf item of 231,525,888 bytes.*

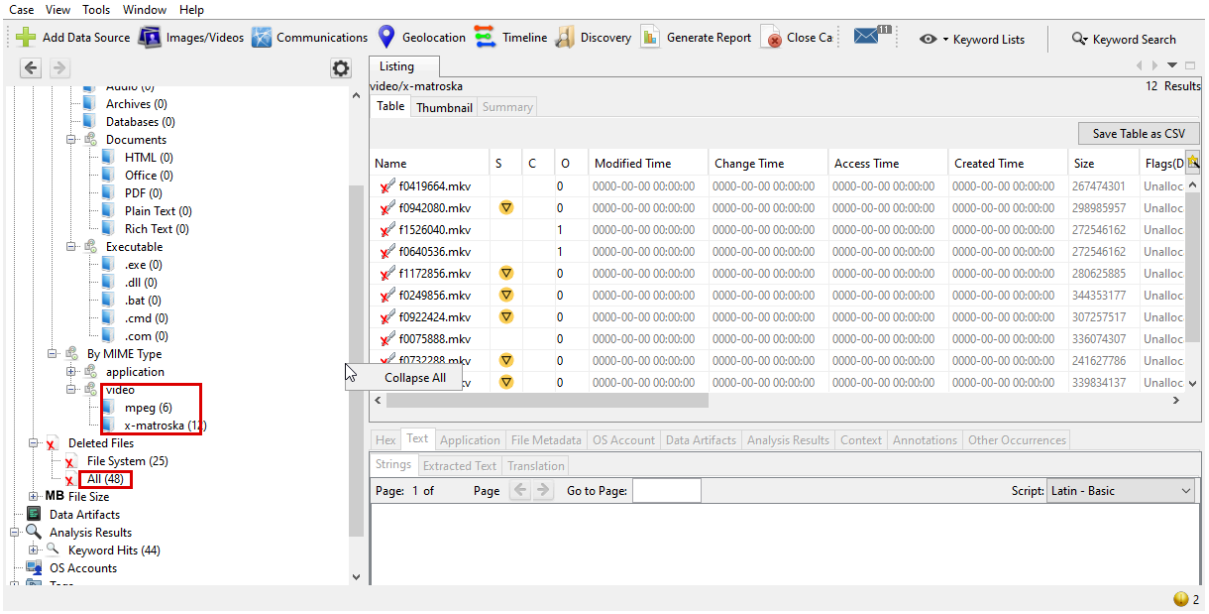


Figure 24: *File Types > By MIME Type > video: mpeg (6) and x-matroska (12), with the x-matroska listing.*

6.6 Keyword search

The Keyword Search module returned 44 hits, all from the built-in "Email Addresses" list; the single literal and single regular-expression lists returned 0 (Figure 25). The addresses themselves were not captured, and no custom keyword list is shown.

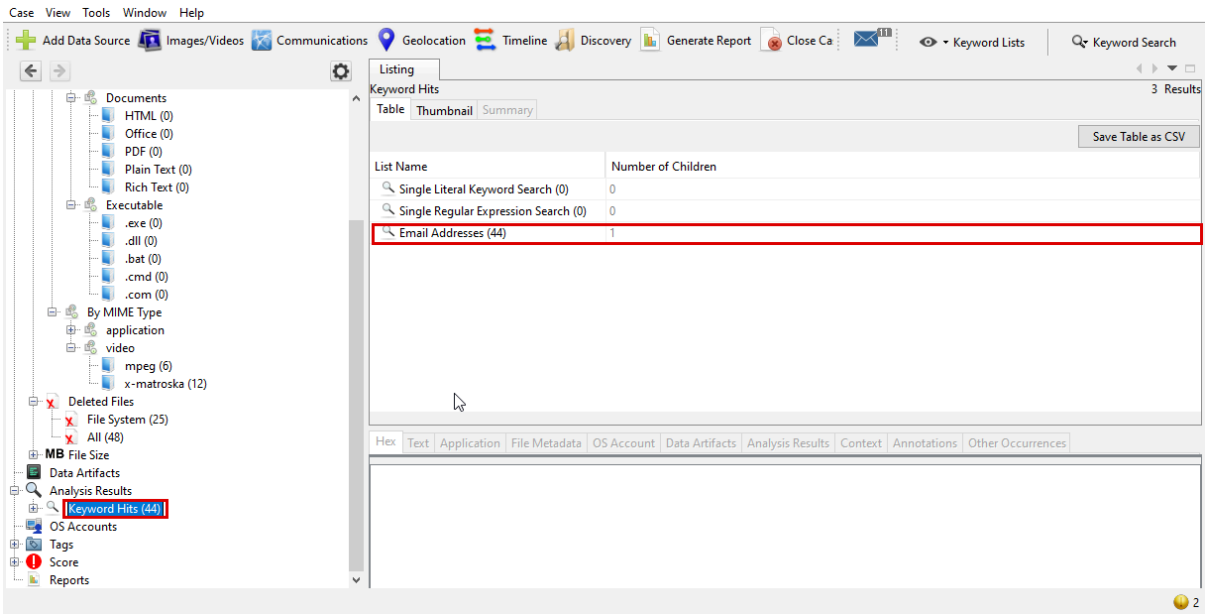


Figure 25: *Keyword Hits (44): Email Addresses (44); literal and regular-expression searches returned 0.*

6.7 Timeline and reporting

The Timeline tool was not used in the captured session. An Autopsy HTML report was generated (Figure 26). Selected timestamps from the deleted-file listing are discussed in Finding F5.

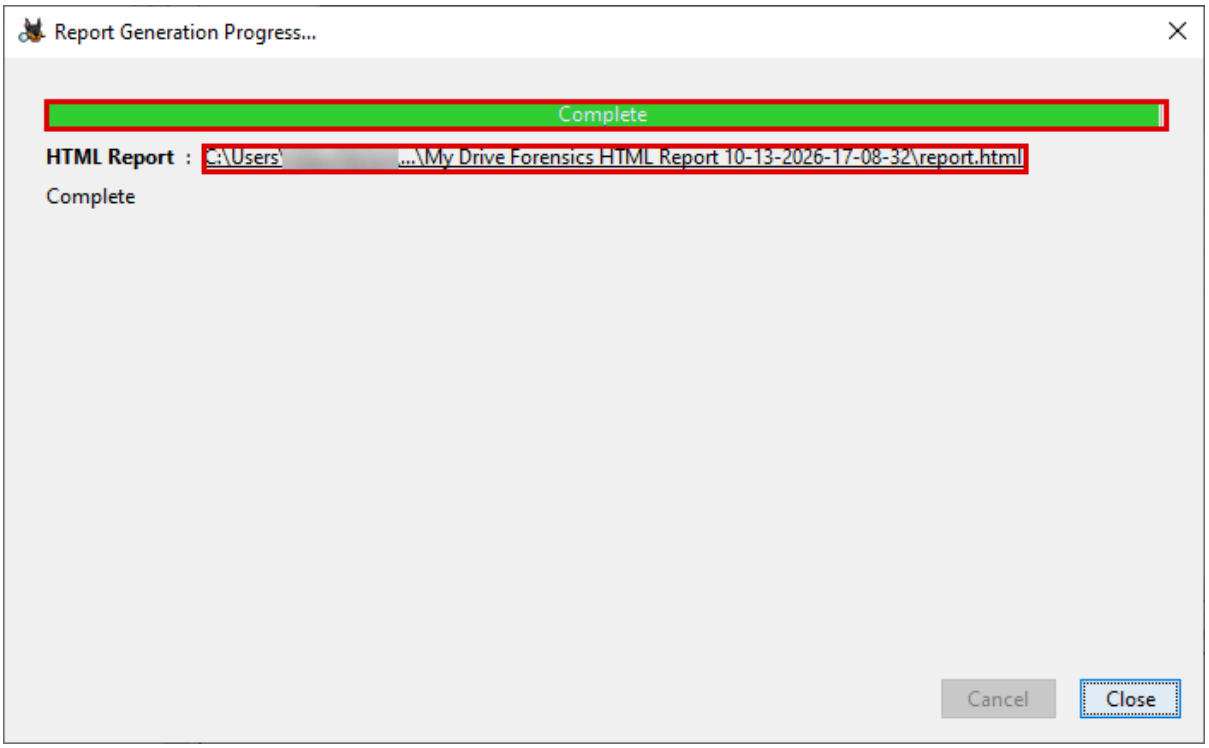


Figure 26: *Autopsy HTML report generation completed (username blurred).*

7. Findings

Each finding gives the observation, the examiner's assessment, and the supporting figures.

F1 The drive was quick-formatted before acquisition

Observation. Windows shows a Quick Format to FAT32 with 4096-byte allocation units and label NO NAME. Free space changed from 6.51 GB to 7.60 GB of 7.60 GB.

Assessment. A quick format rebuilds file-system structures but does not overwrite the data area, so earlier data can remain in unallocated space. The drive's state before the format is unknown.

Evidence: Figure 1, Figure 2

F2 The image was acquired and verified successfully

Observation. E01 image created in 0:09:14 with verification enabled. SHA1 computed, stored and report values are identical (bab4ed750a1e4cb820028fc3b257e19857a8c73b); result Match; no bad blocks.

Assessment. The image is a verified copy of the read data. The match proves the image was not altered after creation, but not that the source was unmodified during reading (no write blocker was used).

Evidence: Figures 9 to 12

F3 The image shows a FAT-style volume with recoverable regions

Observation. MyDrive.E01 contains $OrphanFiles (15), $CarvedFiles (23 items in folder 1), $Unalloc (8) and System Volume Information (4).

Assessment. Autopsy recognised file-system structures after the format. The contents of System Volume Information were not examined.

Evidence: Figure 19, Figure 20

F4 48 deleted items remain after the format

Observation. Deleted Files > All shows 48 items: 25 file-system entries and 23 other items, matching the 23 items under $CarvedFiles/1.

Assessment. Data survives a quick format. The 23 carved items have no file-system metadata (names are generated by the carving process).

Evidence: Figure 21, Figure 19

F5 Deleted file-system entries carry dates in 2025, plus one anomalous entry

Observation. Entries 000~1.TS, 000~1.DVR and INFO3~1.DVR appear in two groups, created 2025-05-16 21:04:04 and 22:03:48 UTC. The .TS entries were modified at 21:23:20 and 23:38:52 UTC. Entries "_TS_20~1.31-", [current folder] and [parent folder] carry 2025-04-22 00:01:32 UTC. An entry named "^m^^D^^^.oo&" shows modified 1980-08-02, accessed 1980-02-07 and created 1981-04-01. Only the first ten of 48 rows are visible.

Assessment. The names and extensions (.TS, .DVR) are consistent with recorder-style video output, but this is not confirmed because content was not reviewed. The 1980/1981 dates and garbled name are consistent with a corrupted or misread directory entry and should not be treated as reliable.

Evidence: Figure 21, Figure 22

F6 Video files were recovered from unallocated space

Observation. 18 files are video by MIME type (12 x-matroska, 6 mpeg). The visible x-matroska items are named in the carved pattern, flagged unallocated, have zero timestamps and range from about 241 MB to 836 MB. The four .mpg items are each exactly 65,536 bytes.

Assessment. Substantial video data remains on the drive after the format. Carved files lack names and dates, so their origin and time cannot be established from metadata. Their content is unverified.

Evidence: Figure 19, Figure 23, Figure 24

F7 Recovered data may predate the most recent file set (inference)

Observation. The eight visible carved .mkv files total about 2.56 GB (sum of their listed byte sizes), whereas about 1.09 GB was in use before the format.

Assessment. This suggests some recovered content comes from earlier use of the drive rather than the last set of files. This is an inference only: carved file sizes can be over-estimated and may overlap, so it needs verification by hashing and content review.

Evidence: Figure 19, Figure 1

F8 Built-in keyword search found 44 email-address hits

Observation. Keyword Hits shows Email Addresses (44); literal and regex lists returned 0.

Assessment. The hits indicate text resembling email addresses exists in the image. Their source files and relevance were not examined.

Evidence: Figure 25

F9 One item requires header verification

Observation. The extension view lists f1409721.swf at 231,525,888 bytes among the videos. Its MIME type was not captured, and the MIME video list shows only mpeg and x-matroska.

Assessment. A .swf (Flash) file of this size is unusual, but a signature mismatch can be neither confirmed nor excluded from the screenshots.

Evidence: Figure 23, Figure 24

8. Limitations

- The drive was formatted before acquisition, so its original contents and state cannot be determined.
- Recovered and carved files were not opened, hashed or played, so no statement is made about their content, authorship or lawfulness.
- Carved files have no names or timestamps, and carved sizes may be inaccurate.
- Date and time of the format, imaging and analysis were not captured, except the Autopsy report time.
- **No write blocker was used** during acquisition, so the source device was exposed to the host system and could have been altered by it.
- The pendrive's **model and serial number were not recorded**, so the device cannot be uniquely identified beyond its reported make, capacity and the name shown by FTK Imager.
- **MD5 values were not recorded**; only the SHA1 values visible in the verification window are available.
- Only a selection of Autopsy views was captured: no timeline, signature-mismatch view, hash-lookup results or the email-hit contents.
- Some table columns (e.g. sizes of the .TS, .DVR files) were truncated in the screenshots.
- The make "SanDisk" is examiner-reported and differs from the controller name shown by FTK Imager.

9. Conclusion and Recommendations

The drive was imaged to a verified E01 file and examined in Autopsy. The evidence shows that quick formatting did not remove earlier data: 48 deleted items, unallocated-space segments and 18 video files by MIME type remain recoverable. Beyond this, the findings are limited to what is visible in the figures; the content, origin and purpose of the recovered data are undetermined.

Recommendations (each follows from the limitations above):

1. Export the recovered and carved files, record MD5/SHA1 for each, and review them in an isolated environment.
2. Review the context of the 44 email-address hits.
3. Verify the file header of f1409721.swf and run the signature-mismatch view.

4. Use a write blocker, and record the device model and serial number, MD5 and SHA1 values, tool versions and the date and time of each step.
5. In future cases, acquire the device before any modification (including formatting) and document the handling.

Appendix A: Hash Log and Tool Settings

Item	Algorithm	Value
E01 image: computed hash	SHA1	bab4ed750a1e4cb820028fc3b257e19857a8c73b
E01 image: stored verification hash	SHA1	bab4ed750a1e4cb820028fc3b257e19857a8c73b
E01 image: report hash	SHA1	bab4ed750a1e4cb820028fc3b257e19857a8c73b
E01 image	MD5	Not recorded

SHA1 values were transcribed from Figure 12 and should be checked against the FTK Imager text report.

Setting	Value
FTK: image type / fragment / compression	E01 / 1500 MB / 6
FTK: verify after creation	Ticked
FTK: precalculate progress statistics	Not ticked
FTK: directory listings after creation	Not ticked
FTK: AD encryption	Not used
Autopsy: case type	Single-user
Autopsy: time zone / sector size	UTC (GMT+0:00) / Auto detect
Autopsy: ignore orphan files in FAT	Not ticked
Autopsy: ingest context	All Files, Directories, and Unallocated Space
Autopsy: modules ticked (visible)	Embedded File Extractor, Picture Analyzer, Keyword Search, Email Parser, Encryption Detection, Interesting Files Identifier, Central Repository, PhotoRec Carver, Virtual Machine Extractor, Data Source Integrity

Appendix B: Supplementary Screenshots

These screenshots document intermediate steps and progress states.

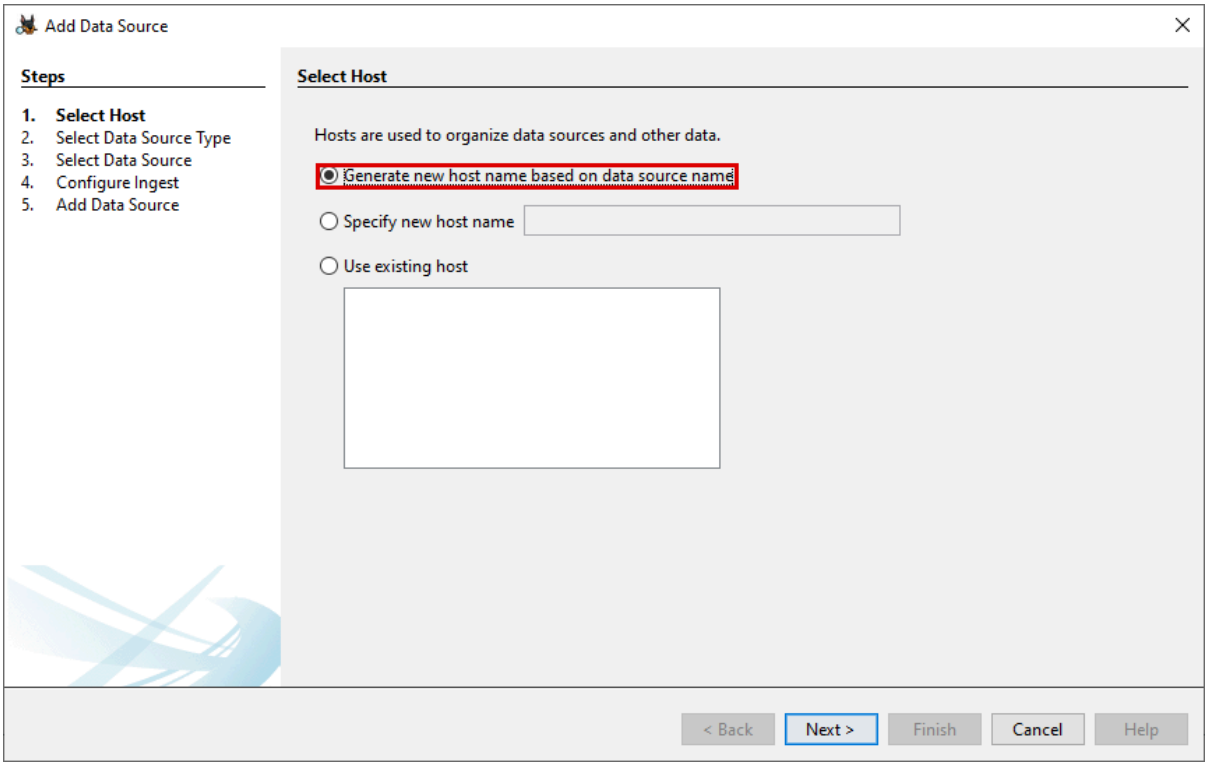


Figure 27: *Autopsy, Select Host: host name generated from the data source name.*



Figure 28: *Autopsy, Creating Case: case database being created.*

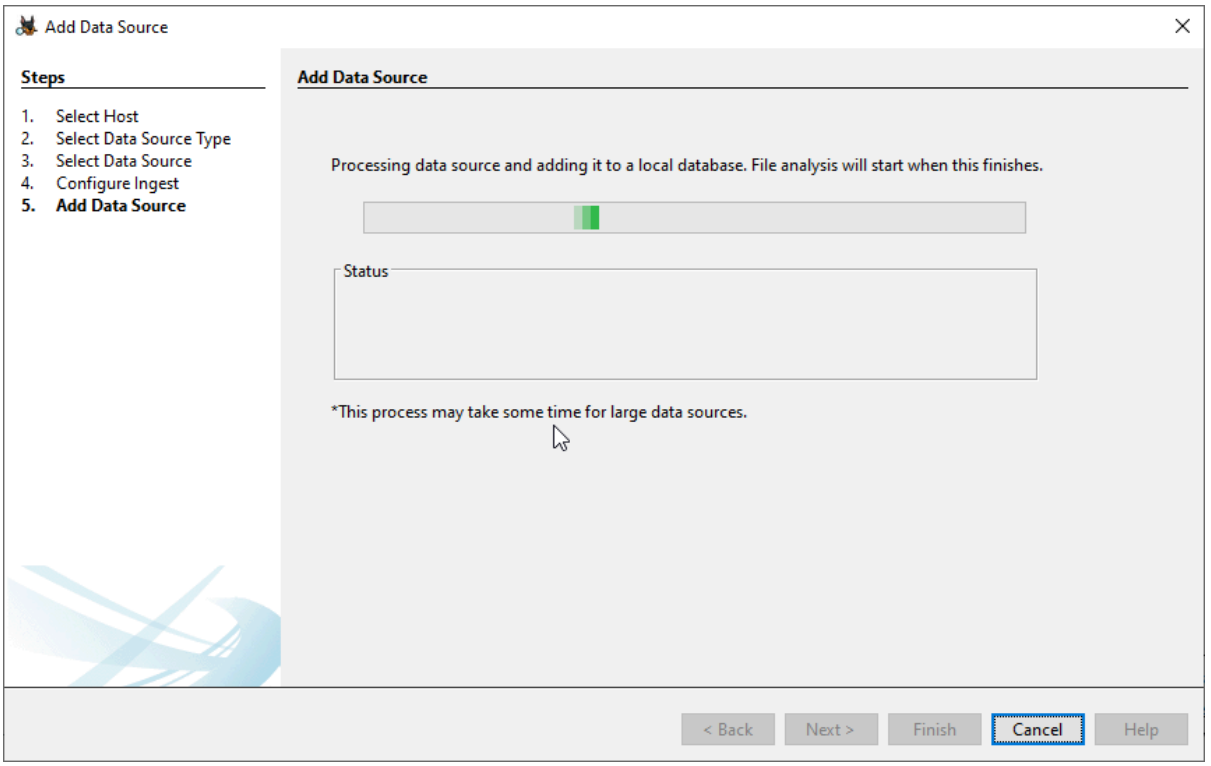


Figure 29: *Autopsy, Add Data Source: data source being processed into the local database.*

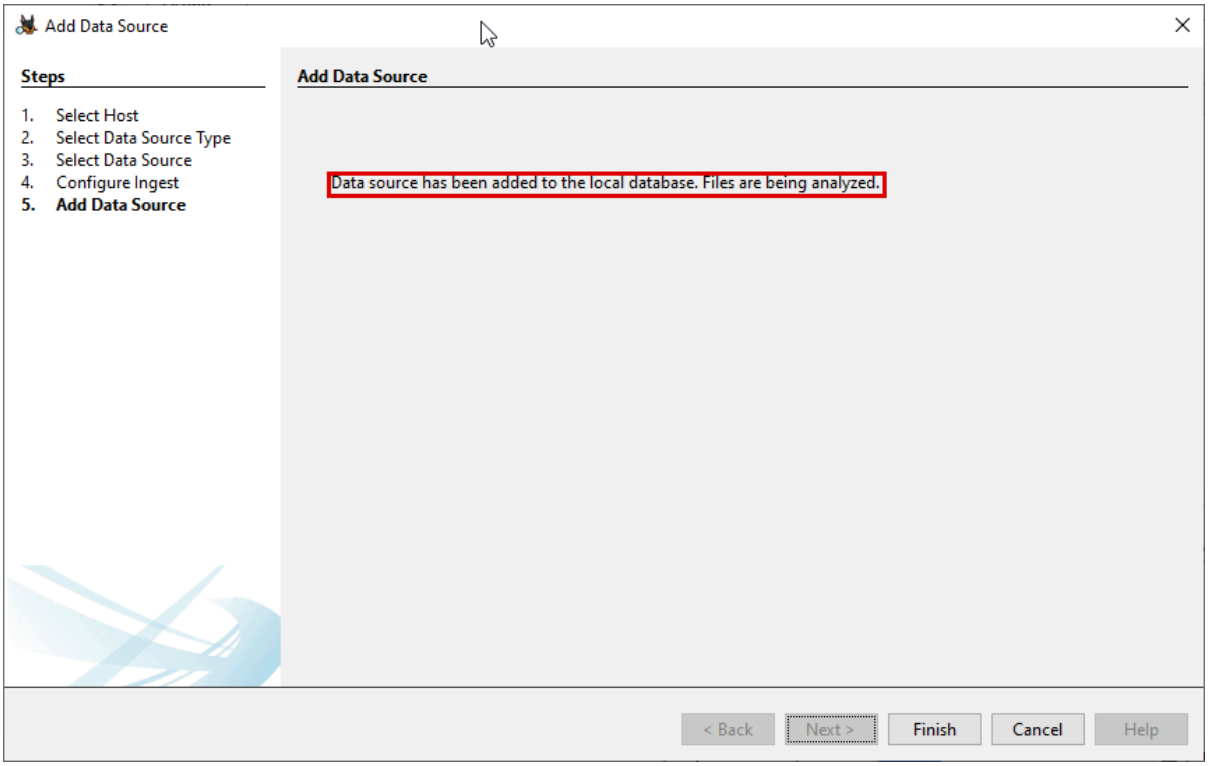


Figure 30: *Autopsy, Add Data Source: data source added; files being analysed.*

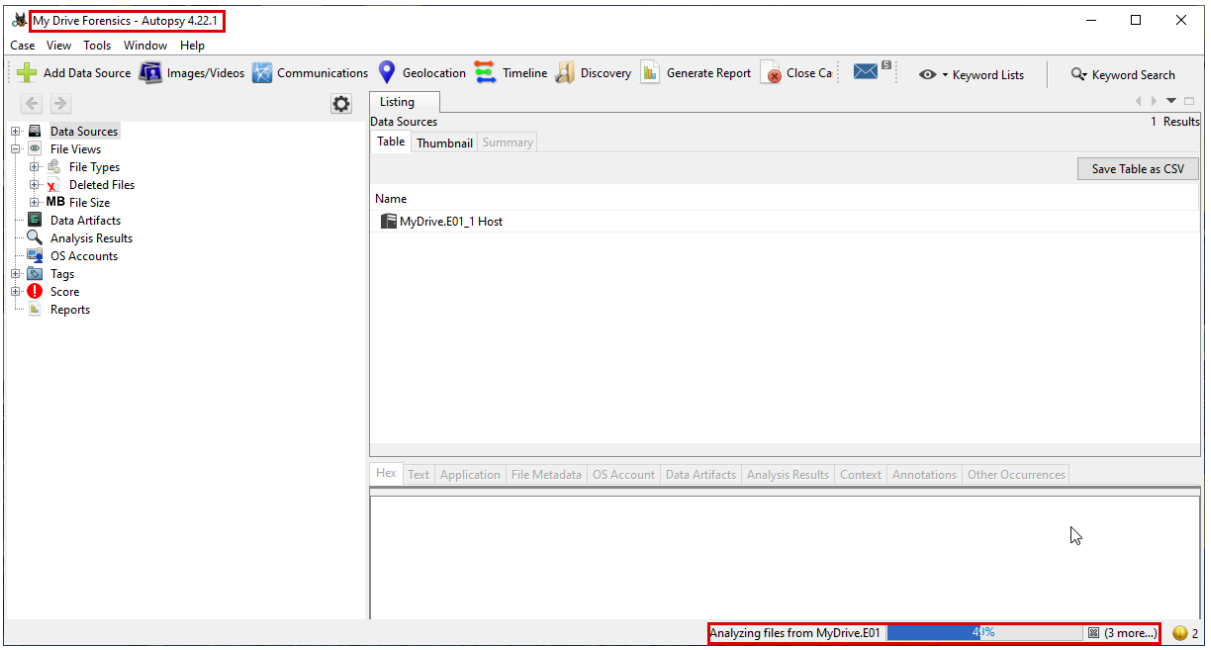


Figure 31: *Autopsy main window during ingest (Analyzing files from MyDrive.E01, 49%).*

Appendix C: References

- Carrier, B. (2005) File System Forensic Analysis. Addison-Wesley.
- International Organization for Standardization (2012) ISO/IEC 27037:2012 Guidelines for identification, collection, acquisition and preservation of digital evidence.
- Kent, K., Chevalier, S., Grance, T. and Dang, H. (2006) NIST Special Publication 800-86: Guide to Integrating Forensic Techniques into Incident Response.
- Autopsy and The Sleuth Kit documentation. Available at: https://www.sleuthkit.org/autopsy/docs.php
- Exterro. FTK Imager User Guide.